

JEE MAIN REVISION MASTER: COORDINATION COMPOUNDS 2026 4 QUESTIONS 6.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Coordination Compounds - Core Application	32 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Coordination Compounds - Core Application (Very High Yield — 32 PYQs)

Core Principle: Key problem pattern in Kcet Coordination Compounds.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Match List-I with List-II List-I (Complex) List-II (Geometry) (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (i) Trigonal bipyramidal (b) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (ii) Octahedral (c) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (iii) Tetrahedral (d) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (iv) Square planar" [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Given below are two statements: Statement-I: The M-C σ bond is formed by the donation of lone pair of electrons on the carbon atom. Statement-II: The M-C π bond is formed by the back donation of electrons from the metal d-orbital to the vacant π^* orbital of the carbon atom." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "How many ions per molecule are produced from the complex $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$ in solution?" [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match List-I with List-II List-I (Complex) List-II (Geometry) (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (i) Trigonal bipyramidal (b) $[\text{NiCl}_4]^{2-}$ (ii) Octahedral (c) $[\text{Ni}(\text{CN})_4]^{2-}$ (iii) Tetrahedral (d) $[\text{Fe}(\text{CO})_5]$ (iv) Square planar Choose the correct answer from the options given below.

- (A) a-ii, b-iii, c-iv, d-i
- (B) a-ii, b-i, c-iii, d-iv
- (C) a-iii, b-ii, c-iv, d-i
- (D) a-i, b-iii, c-iv, d-ii

Correct Answer: None

Solution Analysis: For each complex, we identify the coordination number and the usual geometry. $[\text{Co}(\text{NH}_3)_6]^{3+}$ has 6 ligands around cobalt. So its coordination number is 6, and complexes with coordination number 6 generally have octahedral geometry. (a) $\rightarrow$ (ii) $[\text{NiCl}_4]^{2-}$ has 4 chloride ligands around nickel. Here nickel is in +2 oxidation state, so it is a d^8 ion. Since Cl^- is a weak field ligand, the complex remains tetrahedral. (b) $\rightarrow$ (iii) $[\text{Ni}(\text{CN})_4]^{2-}$ also has nickel in +2 oxidation state, so it is again d^8 . B